

**Boolean Function:** an expression built from binary variables, the constants 0 and 1, and the logic operators AND, OR, NOT. Boolean Functions are denoted with an “F”

Ex)  $F = x + y'z$  is in the boolean function format.

**Truth Tables:** list the output for every possible combination of inputs. Let's look at the truth table for  $F = x + y'z$

| $x$ | $y$ | $z$ | $F$ |
|-----|-----|-----|-----|
| 0   | 0   | 0   | 0   |
| 0   | 0   | 1   | 1   |
| 0   | 1   | 0   | 0   |
| 0   | 1   | 1   | 0   |
| 1   | 0   | 0   | 1   |
| 1   | 0   | 1   | 1   |
| 1   | 1   | 0   | 1   |
| 1   | 1   | 1   | 1   |

Notice how the output column is now labeled “F” instead of the entire boolean equation. This is the standard for how we will write function truth tables.

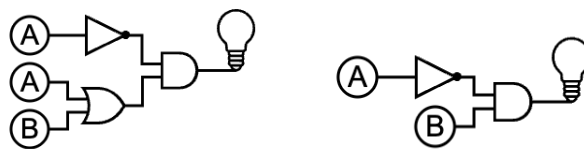
A truth table is unique but an expression is not. This is because the same function can be written in many ways (complicated, simplified) and each way requires a different circuit to be built. What matters when simplifying an expression are the literals.

**Literal:** A single variable in a term, with or without a prime. Think of literals as the wires into each logic gate. Examples of individual literals include  $x$ ,  $x'$ , and  $y$ . The term  $xyz$  has three literals.

Ex) Simplify the following expression to the minimum number of literals.

$$A'(A + B):$$

$$\begin{aligned}
 A'(A + B) &= A'A + A'B && \text{distributive (multiply } A' \text{ in)} \\
 &= 0 + A'B && \text{complement: } A'A = 0 \\
 &= A'B. && \text{identity: } 0 + (\text{anything}) = \text{itself}
 \end{aligned}$$



We saw that one function can be written many different ways. That is helpful for simplifying, but unhelpful when we want one agreed-upon answer. A canonical form fixes this.

**Canonical Form:** a standard way to write a function that you can read straight off the truth table, with no guessing, so everyone arrives at the same expression. There are two forms:

1. **Minterms:** a product (AND) of all variables in a function where the term equals 1 for every one row in the truth table.
  - Complement variables that are 0.
  - Leave variables that are 1 as is.
2. **Maxterms:** a sum (OR) of all variables in a function where the term equals 0 for every one row in the truth table.
  - Complement variables that are 1.
  - Leave variables that are 0 as is.

| $x$ | $y$ | $z$ | minterm  | symbol | maxterm        | symbol |
|-----|-----|-----|----------|--------|----------------|--------|
| 0   | 0   | 0   | $x'y'z'$ | $m_0$  | $x + y + z$    | $M_0$  |
| 0   | 0   | 1   | $x'y'z$  | $m_1$  | $x + y + z'$   | $M_1$  |
| 0   | 1   | 0   | $x'yz'$  | $m_2$  | $x + y' + z$   | $M_2$  |
| 0   | 1   | 1   | $x'yz$   | $m_3$  | $x + y' + z'$  | $M_3$  |
| 1   | 0   | 0   | $xy'z'$  | $m_4$  | $x' + y + z$   | $M_4$  |
| 1   | 0   | 1   | $xy'z$   | $m_5$  | $x' + y + z'$  | $M_5$  |
| 1   | 1   | 0   | $xyz'$   | $m_6$  | $x' + y' + z$  | $M_6$  |
| 1   | 1   | 1   | $xyz$    | $m_7$  | $x' + y' + z'$ | $M_7$  |

Here is a reference table that will help us express our minterms and maxterms based on the inputs that exist. We shouldn't memorize this, but instead understand how they both work using the truth table and examples below.

Let's use the truth table from earlier to determine the minterms and maxterms. We have 3 variables: x, y, z. The rows have specific numbers, the very top row is 0 and the very bottom row is 7.

| $x$ | $y$ | $z$ | $F$ |                                                                                                                                                                                                                                    |
|-----|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | 0   | 0   | 0   | Minterms: the rows that have 1 as the output are 1, 4, 5, 6, 7.<br>$x'y'z + xy'z' + xy'z + xyz' + xyz$ is the function in canonical minterm form.<br>Each term is a product of all the variables, every term is connected using OR |
| 0   | 0   | 1   | 1   |                                                                                                                                                                                                                                    |
| 0   | 1   | 0   | 0   |                                                                                                                                                                                                                                    |
| 0   | 1   | 1   | 0   |                                                                                                                                                                                                                                    |
| 1   | 0   | 0   | 1   | Maxterms: the rows that have 0 as the output are 0, 2, 3<br>$(x+y+z)(x+y'+z)(x+y'+z')$ is the function in canonical maxterm form.<br>Each term is a sum of all the variables, every term is connected using AND                    |
| 1   | 0   | 1   | 1   |                                                                                                                                                                                                                                    |
| 1   | 1   | 0   | 1   |                                                                                                                                                                                                                                    |
| 1   | 1   | 1   | 1   |                                                                                                                                                                                                                                    |

Writing out the full minterms and maxterms can get tiring, so we can express them in a shorthand notation by listing out their associated rows:

Minterms can be shorthand using the rows where 1 is the output:  $\sum (1, 4, 5, 6, 7)$

Maxterms can be shorthand using the rows where 0 is the output:  $\prod (0, 2, 3)$

The minterm list and the maxterm list are just complements of each other (the rows are DISJOINT). To switch forms, swap  $\sum$  and  $\prod$ , and list the numbers you did NOT use.

Ex) Using  $\sum (0, 1, 3, 9)$  and considering that the last row in a truth table is row 10, find the maxterms.

Maxterms are  $\prod (2, 4, 5, 6, 7, 8, 10)$